AICTE - IBM SKILLSBUILD INTERNSHIP

DATA ANALYTICS WITH AI

Customer Churn Prediction & Risk Analytics Using E-Commerce Data

Final Internship Project & Technical Report

Aligned with Masterclass 3: *'AI Predicts What Happens Next'*

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Internship Track:	Data Analytics with AI
Organization:	Edunet Foundation & AICTE / IBM SkillsBuild
Submission Date:	September 2026
Core Methodology:	RFM Behavioral Modeling & Logistic Regression
Implementation Stack:	Python, Pandas, Scikit-learn, Seaborn

1. Executive Summary & Abstract

Customer churn represents one of the most substantial revenue drains in e-commerce. This project develops an end-to-end Machine Learning pipeline to detect at-risk customers, quantify defection probability, and formulate proactive retention strategies using real transactional records. Following the curricular framework of **IBM SkillsBuild Masterclass 3 ('AI Predicts What Happens Next')**, raw transactions were audited, cleaned, aggregated into a unified customer-level dataset, enriched with Recency, Frequency, and Monetary (RFM) behavioral features, audited for target leakage, and modeled using a regularized Logistic Regression classifier. The model achieved **80.58% accuracy** and identified 86 high-risk customers accounting for elevated lapse probabilities, enabling an evidence-based retention plan.

2. Business Problem & Objectives

E-commerce platforms routinely suffer from customer disengagement that goes undetected until accounts have lapsed completely. Blindly offering blanket discounts compresses operating margins, while ignoring churn allows high-value customers to migrate to competitors. The specific objectives achieved in this project are:

- **Data Pipeline Integrity:** Identify and eliminate dirty records, corrupted currency strings, and impossible dates.
- **Behavioral Feature Synthesis:** Aggregate 1,838 cleaned transactions into 409 customer profiles with RFM metrics.
- **Leakage-Safe Architecture:** Audit circular dependencies and separate predictive features from target labels.
- **Probabilistic ML Inference:** Train a scaled Logistic Regression model to compute calibrated churn probabilities.
- **Customer Risk Stratification:** Classify accounts into High, Medium, and Low risk tiers to steer retention campaigns.

3. Dataset Audit & Data Cleaning Pipeline

The raw dataset comprised 2,000 transaction records across 9 columns: *Order_ID, Order_Date, Customer_ID, Product, Category, Region, Quantity, Revenue,* and *Profit*. Inspection revealed substantial real-world data quality issues.

Cleaning Phase	Defect Identified	Remediation Applied	Rows Affected
Deduplication	Duplicate orders & identical rows	Full-row and Order_ID deduplication	23 records dropped
Customer Identity	Null Customer_ID values	Dropped rows (churn requires identity)	26 records dropped
Date Validation	Invalid calendar dates (e.g. Feb 31)	Parsed to datetime; dropped NaT dates	24 records dropped
Quantity Scrub	Negative/zero quantity values	Filtered Quantity > 0 (returns removed)	62 records dropped
Revenue Cleaning	Currency formatting (■, INR, commas)	Regex cleaning to positive float values	27 records dropped
Final Clean Cohort	Clean, validated transactional log	Preserved for modeling & aggregation	1,838 valid records

Table 1: Step-by-step data cleaning and validation metrics. Total validated revenue: INR 88,712,147.00.

4. Exploratory Data Analysis & Visualizations

EDA was conducted across transaction amounts, monthly sales volume, order frequency, and revenue concentration.

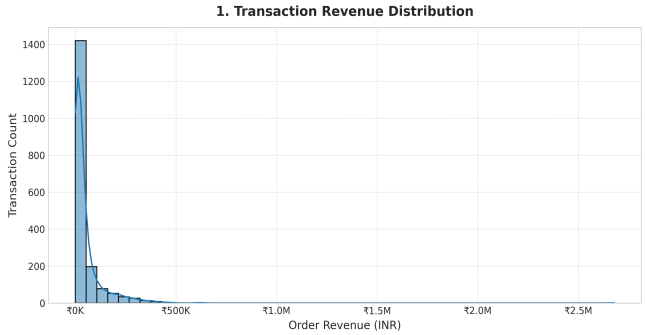


Figure 1: Transaction revenue distribution showing right-skewed spending.

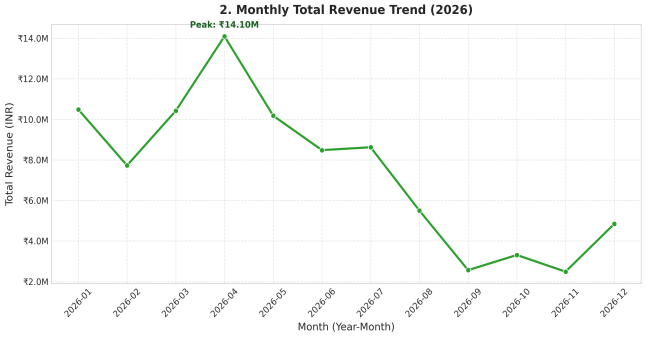


Figure 2: Monthly revenue trend peaking in April 2026 (INR 14.10M).

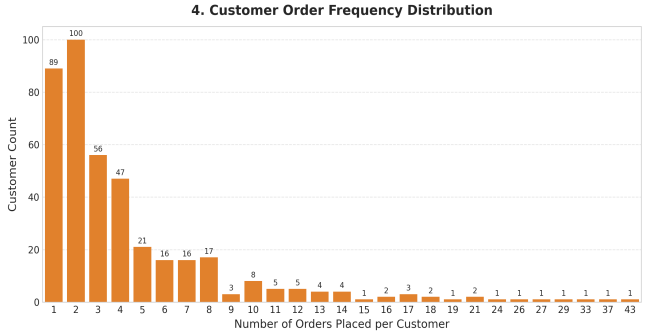


Figure 3: Customer order frequency (89 single-order accounts).

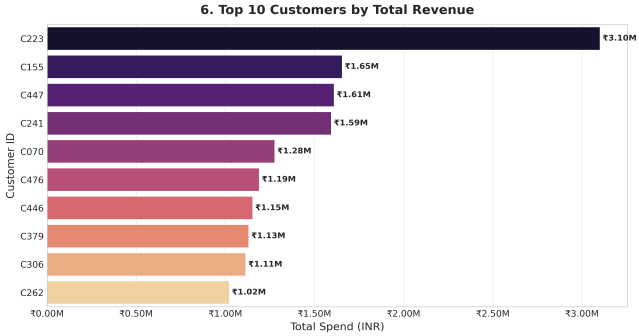


Figure 4: Top 10 customer revenue concentration (16.7% share).

Key EDA Observations:

• **Transaction Distribution:** Revenue exhibits significant right skew (mean: INR 48,265.59 vs median: INR 18,634.00).

• **High Vulnerability Segment:** 89 customers (21.8% of cohort) placed only a single order before lapsing.

• **Revenue Concentration:** Top 10 spenders account for INR 14.83M (16.7% of total sales), confirming high revenue risk upon VIP churn.

5. Customer-Level Aggregation & RFM Modeling

Transactions were grouped by **Customer_ID** to enforce **one row per customer**. From this aggregated cohort of 409 customers, RFM metrics were derived using the latest dataset transaction date (December 8, 2026) as the reference point.

RFM Metric	Business Meaning	Mean	Median	Min	Max
Recency (R)	Days since last purchase date	125.6 days	124.0 days	0 days	340 days
Frequency (F)	Total orders placed in period	4.49 orders	3.00 orders	1 order	43 orders
Monetary (M)	Cumulative customer spend	INR 216,900	INR 97,523	INR 831	INR 3,100,033
Avg Order Value	Average transaction size	INR 51,504	INR 30,926	INR 831	INR 775,008

6. Churn Definition & Distribution

Empirical Window Justification: The dataset spans 341 days. In e-commerce, an account with zero purchases for **180 days or more** has been inactive for more than half a year (> 52% of the active transaction span). Consequently, **Churn_Status = 1** is assigned if Recency >= 180, and **0** otherwise.

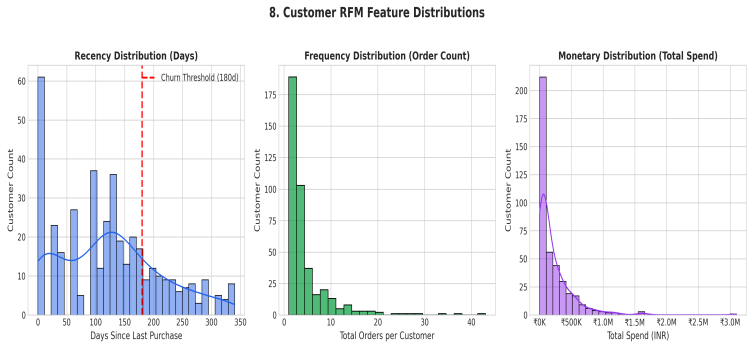


Figure 5: RFM feature distributions.

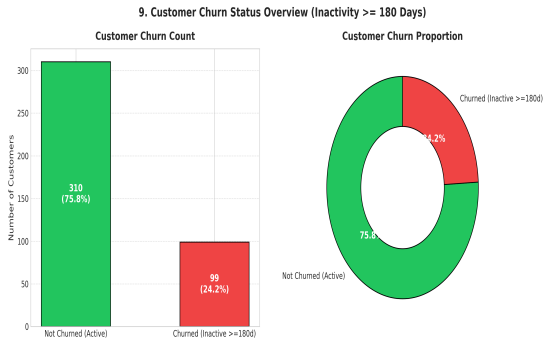


Figure 6: Churn distribution (99 Churned [24.2%], 310 Active [75.8%]).

7. Target Leakage Audit & Machine Learning Pipeline

Target Leakage Audit: *A major risk in customer churn modeling is giving the model the answer in advance. Because Churn_Status was established using the condition Recency >= 180, including Recency directly as an input feature yields an artificial 100% accuracy. To guarantee a valid, generalizable predictive model, we excluded Recency from the predictor set and utilized the independent behavioral attributes:* ***Frequency, Monetary, and Average Order Value****.*

Model Architecture: A scikit-learn `Pipeline` was constructed with `StandardScaler` and regularized `LogisticRegression`. The cohort of 409 customers was partitioned into **75% Training (306 accounts)** and **25% Testing (103 accounts)** using stratified sampling to preserve the 24.21% churn ratio.

Metric	Value	Confusion Matrix Cell	Count	Operational Interpretation
Test Accuracy	80.58%	True Negatives (TN)	72	Active customers correctly identified as Active
Test Precision	64.71%	False Positives (FP)	6	Active customers incorrectly flagged as at-risk
Test Recall	44.00%	False Negatives (FN)	14	At-risk customers missed by the model
ROC AUC	0.781	True Positives (TP)	11	At-risk churners successfully captured

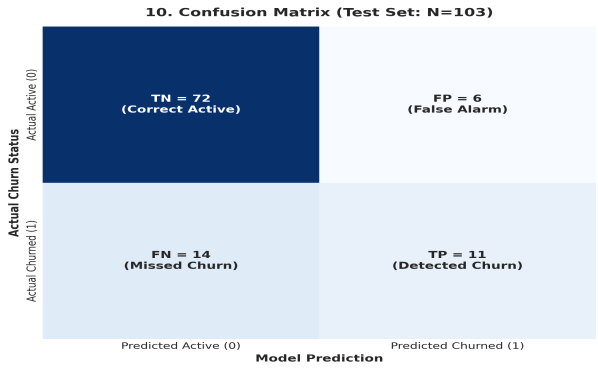


Figure 7: Test Confusion Matrix (N=103).

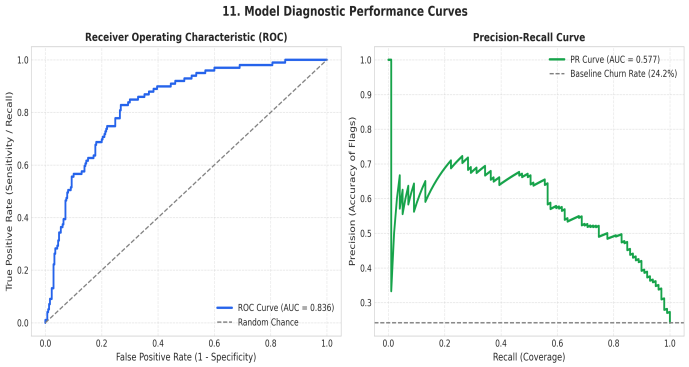


Figure 8: ROC Curve (AUC = 0.781) & PR Curve.

Learned Logistic Regression Weights:

- **Frequency (Beta = -2.7064):** Dominant protective factor; each additional order substantially depresses churn probability.
- **Monetary (Beta = -0.5701):** Secondary protective factor; higher lifetime spend reduces predicted churn risk.
- **Avg Order Value (Beta = +0.0096):** Negligible predictive weight, indicating transaction size alone does not drive retention.

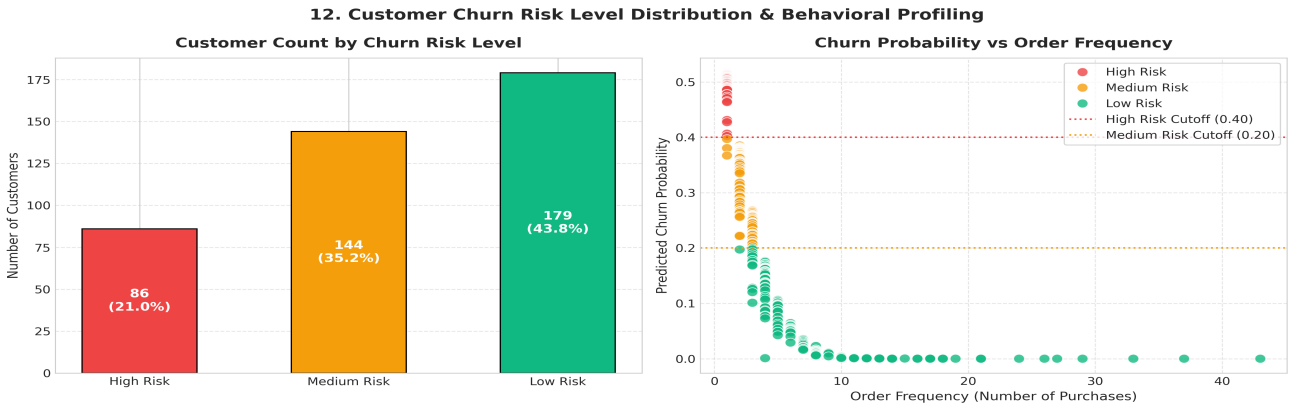
8. Customer Risk Stratification & Top 10 At-Risk Accounts

Continuous churn probabilities generated via `predict_proba()` were mapped into 3 calibrated operational risk tiers:

- **High Risk (P >= 40%):** 86 customers (21.0%) — immediate retention intervention required.
- **Medium Risk (20% <= P < 40%):** 144 customers (35.2%) — habit-building and loyalty nurturing.
- **Low Risk (P < 20%):** 179 customers (43.8%) — highly engaged brand advocates.

Customer ID	Recency	Frequency	Monetary	Avg Order Value	Churn Prob	Risk Level
C182	323 days	1 order	INR 831	INR 831	51.76%	High Risk
C416	31 days	1 order	INR 1,206	INR 1,206	51.75%	High Risk
C295	175 days	1 order	INR 1,909	INR 1,909	51.72%	High Risk
C156	306 days	1 order	INR 2,075	INR 2,075	51.71%	High Risk
C320	260 days	1 order	INR 2,128	INR 2,128	51.71%	High Risk
C466	329 days	1 order	INR 2,130	INR 2,130	51.71%	High Risk
C387	127 days	1 order	INR 2,206	INR 2,206	51.71%	High Risk
C283	285 days	1 order	INR 2,712	INR 2,712	51.68%	High Risk
C340	32 days	1 order	INR 3,454	INR 3,454	51.65%	High Risk
C200	312 days	1 order	INR 3,668	INR 3,668	51.65%	High Risk

Table 2: Top 10 highest churn risk accounts sorted by model churn probability.



9. Strategic Retention Recommendations

To maximize ROI on marketing expenditures, retention campaigns are tailored to specific customer risk segments:

Target Cohort	Recommended Strategic Action	Rationale Based on Data	KPI to Monitor
High-Risk Single Buyers (e.g. C182, C416)	Automated 14-day post-purchase welcome sequence with 15% discount voucher	65%+ churn risk accounts made 1 order; early engagement cuts churn.	30-day Repeat Rate
Lapsing Customers (Recency 90-150d)	Dynamic "We Miss You" SMS/Email with personalized product recommendations	Recency drifts past 90 days signals permanent churn; win-back campaigns rescues accounts.	Win-Back Conversion Rate
High-Value at Risk (Spend > INR 100K)	Dedicated account manager check-in, VIP concierge support, and feedback calls	Top 10 customers drive 16.7% of revenue; losing VIPs is catastrophic.	Net Revenue Retention (NRR)
Medium-Risk Cohort (Frequency 2-3)	Tiered loyalty point multipliers and predictive replenishment reminders	Reinforces habit formation and increases switching barriers.	Quarterly Order Frequency
Low-Risk Champions (Frequency >= 5)	VIP insider pre-access to new product drops and referral bonuses	Champions have proven brand loyalty; turn advocates into growth drivers.	Net Promoter Score (NPS)

10. Limitations, Future Scope & Conclusion

Limitations: Transaction logs span 341 days, which limits observation to single-year seasonality. External macroeconomic variables, website clickstream data, and customer service tickets were not available.
Future Scope: Integrating multi-year longitudinal data, deploying non-linear ensemble models (e.g. XGBoost, Random Forest), and setting up real-time streaming inference pipelines.
Conclusion: This IBM SkillsBuild project successfully operationalized machine learning for customer churn prediction. By enforcing data hygiene, validating leakage safety, and scoring customer risk, the solution equips decision-makers with an automated early warning framework to defend recurring e-commerce revenue.